

Bahir Dar Institute of Technology

Faculty of Mechanical and Industrial Engineering

3rd mechanical Engineering Fluid Mechanics worksheet Chapter 4

1. An idealized velocity field is given by the formula $\mathbf{v} = 4tx \mathbf{i} - 2t^2y \mathbf{j} + 4xz \mathbf{k}$. Is this flow field steady or unsteady? Is it two- or three-dimensional? Compute the acceleration vector at the point $(x, y, z) = (1, 1, 0)$. Is the flow satisfying continuity equation?
2. The velocity components for a certain incompressible, steady-flow field are $u = x^2 + y^2 + z^2$, $v = xy + yz + z$, $w = ?$. Determine the form of the z component, w , required to satisfy the continuity equation.
3. Consider the steady, two-dimensional velocity field given by $\mathbf{V} = (u, v) = (1.6 + 2.8x) \mathbf{i} + (1.5 - 2.8y) \mathbf{j}$. Verify that this flow field is incompressible.
4. An idealized incompressible flow has the proposed three-dimensional velocity distribution
 $\mathbf{V} = 4xy^2 \mathbf{i} + f(y) \mathbf{j} - zy^2 \mathbf{k}$. If this flow exist find $f(y)$?
5. Consider a steady, two-dimensional, incompressible flow of a Newtonian fluid in which the velocity field is known: $u = -2xy$, $v = y^2 - x^2$, $w = 0$. (a) Does this flow satisfy conservation of mass? (b) Find the pressure field, $p(x, y)$ if the pressure at the point $(x=0, y=0)$ is equal to p_a .
6. A two-dimensional incompressible flow is defined by $u = -\frac{Ky}{x^2 + y^2}$ $v = \frac{Kx}{x^2 + y^2}$ Where $K = \text{constant}$. Is this flow irrotational? If so, find Its velocity potential, sketch a few potential lines, and interpret the flow pattern.
7. A two-dimensional incompressible flow field is defined by the velocity components

$$u = 2V \left(\frac{x}{L} - \frac{y}{L} \right) \quad v = -2V \frac{y}{L}$$

Where V and L are constants. If they exist, find the stream function and velocity potential.